

1605.05775: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1605.05775_F00001	1	■ 1.000	$\{ \ }^{\{1,2\}}$	$1,2$	E. Miles Stoudenmire ^{1,2} and David J. Schwab ³
1605.05775_F00002	1	■ 1.000	$\Phi(\mathbf{x})$	$\Phi(\mathbf{x})$	map $\Phi(\mathbf{x})$ before being classified by a decision function
1605.05775_F00003	1	■ 1.000	W	W	The feature vector $\Phi(\mathbf{x})$ and weight vector W can be ex-
1605.05775_F00004	1	■ 1.000	Φ	Φ	ture map Φ , we find the optimal weight vector W can
1605.05775_F00005	2	■ 1.000	d	d	local feature-space dimension d . Finally, we discuss the
1605.05775_F00006	2	■ 1.000	N	N	N independent systems corresponds to taking the tensor
1605.05775_F00007	2	■ 1.000	$\Phi^{s_1 s_2 \cdots s_N}$	$\Phi^{s_1 s_2 \cdots s_N}$	The tensor $\Phi^{s_1 s_2 \cdots s_N}$ is the tensor product of the same
1605.05775_F00008	2	■ 1.000	$\phi^{s_j}(\mathbf{x}_j)$	$\phi^{s_j}(\mathbf{x}_j)$	local feature map $\phi^{s_j}(\mathbf{x}_j)$ applied to each input $\mathbf{x}_j$, where
1605.05775_F00009	2	■ 1.000	$\mathbf{x}_j$	$\mathbf{x}_j$	local feature map $\phi^{s_j}(\mathbf{x}_j)$ applied to each input $\mathbf{x}_j$, where
1605.05775_F00010	2	■ 1.000	s_j	s_j	the indices s_j run from 1 to d ; the value d is known as
1605.05775_F00011	2	■ 1.000	d^N	d^N	in a d^N -dimensional space or as an order- N tensor. The
1605.05775_F00012	2	■ 1.000	j^{th}	j^{th}	for black. If the grayscale pixel value of the j^{th} pixel
1605.05775_F00013	2	■ 1.000	$\mathbf{x}_j \in [0, 1]$	$\mathbf{x}_j \in [0, 1]$	is $\mathbf{x}_j \in [0, 1]$, a simple choice for the local feature map
1605.05775_F00014	2	■ 1.000	ϕ^{s_j}	ϕ^{s_j}	perspective, ϕ^{s_j} is the normalized wavefunction of a sin-
1605.05775_F00015	2	■ 1.000	$d=2$	$d = 2$	FIG. 3. For the case of a grayscale image and $d = 2$, each
1605.05775_F00016	2	■ 1.000	ℓ	ℓ	tion Eq. (4) to a set of functions indexed by a label ℓ
1605.05775_F00017	2	■ 0.984	$\mathbf{x}$	$\mathbf{x}$	and classifying an input $\mathbf{x}$ by choosing the label ℓ for
1605.05775_F00018	2	■ 1.000	$ f^\ell(\mathbf{x}) $	$ f^\ell(\mathbf{x}) $	which $f^\ell(\mathbf{x})$ is largest.
1605.05775_F00019	2	■ 1.000	W^ℓ	W^ℓ	the weight vector W^ℓ . Though one can view W^ℓ as a
1605.05775_F00020	2	■ 1.000	$N+1$	$N + 1$	W^ℓ as an order $N + 1$ tensor where ℓ is a tensor index and
1605.05775_F00021	3	■ 1.000	$f^\ell(\mathbf{x})$	$f^\ell(\mathbf{x})$	input vector $\Phi(\mathbf{x})$ defines the decision function $f^\ell(\mathbf{x})$. The
1605.05775_F00022	3	■ 1.000	$W_{s_1 s_2 \cdots s_N}^\ell$	$W_{s_1 s_2 \cdots s_N}^\ell$	Because the weight tensor $W_{s_1 s_2 \cdots s_N}^\ell$ has $N_L \cdot d^N$ com-
1605.05775_F00023	3	■ 1.000	$N_L \cdot d^N$	$N_L \cdot d^N$	Because the weight tensor $W_{s_1 s_2 \cdots s_N}^\ell$ has $N_L \cdot d^N$ com-
1605.05775_F00024	3	■ 1.000	N_L	N_L	ponents, where N_L is the number of labels, we need a
1605.05775_F00025	3	■ 1.000	α_j	α_j	and is illustrated in Fig. 5. Each “virtual” index α_j has a
1605.05775_F00026	3	■ 1.000	m	m	dimension m which is known as the <i>bond dimension</i> and
1605.05775_F00027	3	■ 1.000	$\{\alpha_j\}$	$\{\alpha_j\}$	over the $\{\alpha_j\}$ indices from left to right via a sequence of
1605.05775_F00028	3	■ 1.000	n	n	where n runs over the N_T training inputs and L_n is the
1605.05775_F00029	3	■ 1.000	N_T	N_T	where n runs over the N_T training inputs and L_n is the
1605.05775_F00030	3	■ 1.000	L_n	L_n	where n runs over the N_T training inputs and L_n is the
1605.05775_F00031	3	■ 1.000	$\delta_{L_n}^\ell = 1$	$\delta_{L_n}^\ell = 1$	$\delta_{L_n}^\ell = 1$ if ℓ equals L_n and zero otherwise, and represents
1605.05775_F00032	3	■ 1.000	f^ℓ	f^ℓ	the ideal output of the function f^ℓ for the input $\mathbf{x}_n$.)
1605.05775_F00033	3	■ 1.000	$\mathbf{x}_n$	$\mathbf{x}_n$	the ideal output of the function f^ℓ for the input $\mathbf{x}_n$.)
1605.05775_F00034	3	■ 1.000	j	j	Say we want to improve the tensors at sites j and $j + 1$
1605.05775_F00035	3	■ 1.000	$j+1$	$j + 1$	Say we want to improve the tensors at sites j and $j + 1$
1605.05775_F00036	3	■ 1.000	$A_{s_j}^\ell$	$A_{s_j}^\ell$	MPS tensors $A_{s_j}^\ell$ and $A_{s_{j+1}}$ into a single “bond tensor”
1605.05775_F00037	3	■ 1.000	$A_{s_{j+1}}$	$A_{s_{j+1}}$	MPS tensors $A_{s_j}^\ell$ and $A_{s_{j+1}}$ into a single “bond tensor”
1605.05775_F00038	4	■ 1.000	B^ℓ	B^ℓ	tensor B^ℓ at bond j : (a) forming the bond tensor; (b) project-
1605.05775_F00039	4	■ 0.881	$B_{s_j s_{j+1}}^{\alpha_{j-1} \alpha_{j+1}}$	$B_{s_j s_{j+1}}^{\alpha_{j-1} \alpha_{j+1}}$	$B_{s_j s_{j+1}}^{\alpha_{j-1} \alpha_{j+1}}$ by contracting over the index α_j as shown in
1605.05775_F00040	4	■ 1.000	C	C	Next we compute the derivative of the cost function C

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1605.05775_F00041	4	■ 0.995	$\tilde{\Phi}_n$	$\tilde{\Phi}_n$	projected, four-index version of the input $\tilde{\Phi}_n$ shown on
1605.05775_F00042	4	■ 1.000	$B^{\ell} + \alpha \Delta B^{\ell}$	$B^{\ell} + \alpha \Delta B^{\ell}$	$B^{\ell} + \alpha \Delta B^{\ell}$ as shown in Fig. 7(a), where α is a small
1605.05775_F00043	4	■ 1.000	α	α	$B^{\ell} + \alpha \Delta B^{\ell}$ as shown in Fig. 7(a), where α is a small
1605.05775_F00044	4	■ 0.743	$j+1)$	$j+1)$	$j+1)$. Then we can compute a singular value decom-
1605.05775_F00045	4	■ 0.992	$\left(\alpha_{j-1}, s_j\right)$	(α_{j-1}, s_j)	collective row index (α_{j-1}, s_j) and collective column in-
1605.05775_F00046	4	■ 0.769	$\operatorname{dex}(\ell, \alpha_{j+1}, s_{j+1})$	$\operatorname{dex}(\ell, \alpha_{j+1}, s_{j+1})$	(α_{j+1}, s_{j+1}) as shown in Fig. 7(b). Computing the
1605.05775_F00047	4	■ 1.000	$A_{s_j}^{\prime} = U_{s_j}$	$A_{s_j}^{\prime} = U_{s_j}$	tensor at site j to be $A_{s_j}^{\prime} = U_{s_j}$ and the new tensor at
1605.05775_F00048	4	■ 1.000	$A_{s_{j+1}}^{\prime} = S V_{s_{j+1}}^{\ell}$	$A_{s_{j+1}}^{\prime} = S V_{s_{j+1}}^{\ell}$	site $j+1$ to be $A_{s_{j+1}}^{\prime} = S V_{s_{j+1}}^{\ell}$ where a matrix multipli-
1605.05775_F00049	4	■ 1.000	S	S	at this point, only the m largest singular values in S are
1605.05775_F00050	4	■ 1.000	U	U	sponding columns of U and $V^{\dagger}$) in order to control the
1605.05775_F00051	4	■ 1.000	$V^{\dagger}$	$V^{\dagger}$	sponding columns of U and $V^{\dagger}$) in order to control the
1605.05775_F00052	4	■ 1.000	$A_{s_j}^{\prime}$	$A_{s_j}^{\prime}$	tary matrices similar to the definition of $A_{s_j}^{\prime}$ above, then
1605.05775_F00053	5	■ 1.000	B^{ℓ}	B^{ℓ}	tensor B^{ℓ} , the dimension of the new MPS bond can be
1605.05775_F00054	5	■ 1.000	$d^3 m^3 N N_L N_T$	$d^3 m^3 N N_L N_T$	The scaling of the above algorithm is $d^3 m^3 N N_L N_T$,
1605.05775_F00055	5	■ 0.351	14×14	14×14	down to 14×14 by averaging clusters of four pixels; other-
1605.05775_F00056	5	■ 1.000	$s_1, s_2, \dots, s_N$	$s_1, s_2, \dots, s_N$	$s_1, s_2, \dots, s_N$. We chose a “zig-zag” ordering shown in
1605.05775_F00057	5	■ 1.000	$m=10$	$m=10$	imum MPS bond dimension m . For $m=10$ we found
1605.05775_F00058	5	■ 0.994	$m=20$	$m=20$	both a training and test error of about 5%; for $m=20$
1605.05775_F00059	5	■ 1.000	$m=120$	$m=120$	sion we tried was $m=120$, where after three sweeps we
1605.05775_F00060	5	■ 1.000	$(N=2)$	$(N=2)$	the input space is two-dimensional ($N=2$). The hidden
1605.05775_F00061	5	■ 1.000	$P_A(x_1, x_2)$	$P_A(x_1, x_2)$	tions, $P_A(x_1, x_2)$ and $P_B(x_1, x_2)$, from which we gener-
1605.05775_F00062	5	■ 1.000	$P_B(x_1, x_2)$	$P_B(x_1, x_2)$	tions, $P_A(x_1, x_2)$ and $P_B(x_1, x_2)$, from which we gener-
1605.05775_F00063	5	■ 1.000	A	A	ate training data points labeled A or B respectively. For
1605.05775_F00064	5	■ 1.000	B	B	ate training data points labeled A or B respectively. For
1605.05775_F00065	5	■ 1.000	$x_1 \in [0, 1]$	$x_1 \in [0, 1]$	simplicity we only consider the square region $x_1 \in [0, 1]$
1605.05775_F00066	5	■ 1.000	$x_2 \in [0, 1]$	$x_2 \in [0, 1]$	and $x_2 \in [0, 1]$.
1605.05775_F00067	5	■ 0.986	(x_1, x_2)	(x_1, x_2)	To train the model, each training point (x_1, x_2) is
1605.05775_F00068	5	■ 1.000	$W_{s_1 s_2}^{\ell}$	$W_{s_1 s_2}^{\ell}$	and the full weight tensors $W_{s_1 s_2}^{\ell}$ for $\ell \in \{A, B\}$ are op-
1605.05775_F00069	5	■ 1.000	$\ell \in \{A, B\}$	$\ell \in \{A, B\}$	and the full weight tensors $W_{s_1 s_2}^{\ell}$ for $\ell \in \{A, B\}$ are op-
1605.05775_F00070	5	■ 1.000	s_1	s_1	is the dimension d of the local indices s_1 and s_2 . For the
1605.05775_F00071	5	■ 1.000	s_2	s_2	is the dimension d of the local indices s_1 and s_2 . For the
1605.05775_F00072	5	■ 1.000	$d > 2$	$d > 2$	For $d > 2$ we generalize $\phi^{s_j}(x_j)$ to be a normalized d -
1605.05775_F00073	5	■ 1.000	P_A	P_A	with increasing d , consider the case where P_A and P_B
1605.05775_F00074	5	■ 1.000	P_B	P_B	with increasing d , consider the case where P_A and P_B
1605.05775_F00075	6	■ 1.000	$P_A(x_1, x_2) = P_B(x_1, x_2)$	$P_A(x_1, x_2) = P_B(x_1, x_2)$	$P_A(x_1, x_2) = P_B(x_1, x_2)$. To make a training set, we sam-
1605.05775_F00076	6	■ 1.000	$d=3$	$d=3$	The $d=3$ decision boundary shown in Fig. 10 begins
1605.05775_F00077	6	■ 1.000	$d=6$	$d=6$	$d=6$ there is extreme overfitting. The decision bound-
1605.05775_F00078	6	■ 1.000	$d=10$	$d=10$	with local dimension $d=10$ on 500 sampled points, 250
1605.05775_F00079	6	■ 0.887	$B)$	$B)$	$B)$. The learned model is able to classify every training
1605.05775_F00080	7	■ 1.000	$f^{\ell}(\mathbf{x}) = W^{\ell} \cdot \Phi(\mathbf{x})$	$f^{\ell}(\mathbf{x}) = W^{\ell} \cdot \Phi(\mathbf{x})$	form $f^{\ell}(\mathbf{x}) = W^{\ell} \cdot \Phi(\mathbf{x})$ can be realized when using a
1605.05775_F00081	7	■ 1.000	$f(\mathbf{x})$	$f(\mathbf{x})$	constraint. Then $f(\mathbf{x})$ is a function of the form
1605.05775_F00082	7	■ 1.000	$\{\phi^s(x)\}, s=1,2,\dots,d$	$\{\phi^s(x)\}, s=1,2,\dots,d$	If the functions $\{\phi^s(x)\}, s=1,2,\dots,d$ form a basis for a
1605.05775_F00083	7	■ 1.000	$x \in [0, 1]$	$x \in [0, 1]$	Hilbert space of functions over $x \in [0, 1]$, then the tensor
1605.05775_F00084	7	■ 1.000	$\mathbf{x} \in [0, 1]^{\times N}$	$\mathbf{x} \in [0, 1]^{\times N}$	$\mathbf{x} \in [0, 1]^{\times N}$. Moreover, if the basis $\{\phi^s(x)\}$ is complete,
1605.05775_F00085	7	■ 1.000	$\{\phi^s(x)\}$	$\{\phi^s(x)\}$	$\mathbf{x} \in [0, 1]^{\times N}$. Moreover, if the basis $\{\phi^s(x)\}$ is complete,
1605.05775_F00086	7	■ 1.000	$\phi(x)$	$\phi(x)$	$\phi(x)$,

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1605.05775_F00087	7	■ 1.000	$\hat{\mathbf{x}}$	$\hat{\mathbf{x}}$	Thus if $\hat{\mathbf{x}}$ is a black and white image with pixel values of
1605.05775_F00088	7	■ 1.000	$\hat{x}_j = \{0, 1\}$	$\hat{x}_j = \{0, 1\}$	only $\hat{x}_j = \{0, 1\}$, then $f(\hat{\mathbf{x}})$ is equal to a single component
1605.05775_F00089	7	■ 1.000	$f(\hat{\mathbf{x}})$	$f(\hat{\mathbf{x}})$	only $\hat{x}_j = \{0, 1\}$, then $f(\hat{\mathbf{x}})$ is equal to a single component
1605.05775_F00090	7	■ 1.000	$W_{s_1 s_2 \dots s_N}$	$W_{s_1 s_2 \dots s_N}$	$W_{s_1 s_2 \dots s_N}$ of the weight tensor. Because each of these
1605.05775_F00091	7	■ 0.998	$x_j \in [0, 1], f(\mathbf{x})$	$x_j \in [0, 1], f(\mathbf{x})$	$x_j \in [0, 1], f(\mathbf{x})$ cannot be an arbitrary function over
1605.05775_F00092	7	■ 1.000	$a \cos(\pi/2 x_j) + b \sin(\pi/2 x_j)$	$a \cos(\pi/2 x_j) + b \sin(\pi/2 x_j)$	functional form $a \cos(\pi/2 x_j) + b \sin(\pi/2 x_j)$ where a and
1605.05775_F00093	7	■ 1.000	a	a	functional form $a \cos(\pi/2 x_j) + b \sin(\pi/2 x_j)$ where a and
1605.05775_F00094	7	■ 1.000	b	b	b are constants determined by the (fixed) values of the
1605.05775_F00095	7	■ 1.000	$C^{\alpha_i \ell \alpha_{i+1}}$	$C^{\alpha_i \ell \alpha_{i+1}}$	$C^{\alpha_i \ell \alpha_{i+1}}$ on some bond i and constrain all MPS site ten-
1605.05775_F00096	7	■ 1.000	i	i	$C^{\alpha_i \ell \alpha_{i+1}}$ on some bond i and constrain all MPS site ten-
1605.05775_F00097	7	■ 1.000	$j \leq i$	$j \leq i$	sors to be <i>left orthogonal</i> for sites $j \leq i$ or <i>right orthogonal</i>
1605.05775_F00098	7	■ 1.000	$j \geq i$	$j \geq i$	for sites $j \geq i$. This means W^ℓ has the decomposition
1605.05775_F00099	7	■ 1.000	V	V	as illustrated in Fig. 12(a). To say the U and V tensors
1605.05775_F00100	7	■ 1.000	$U_{\alpha_{j-1} s_j}^{\alpha_j}$	$U_{\alpha_{j-1} s_j}^{\alpha_j}$	trices $U_{\alpha_{j-1} s_j}^{\alpha_j}$ and $V^{\alpha_{j-1} s_j \alpha_j}$ these tensors have the
1605.05775_F00101	7	■ 1.000	$V^{\alpha_{j-1} s_j \alpha_j}$	$V^{\alpha_{j-1} s_j \alpha_j}$	trices $U_{\alpha_{j-1} s_j}^{\alpha_j}$ and $V^{\alpha_{j-1} s_j \alpha_j}$ these tensors have the
1605.05775_F00102	7	■ 1.000	$U^\dagger U = I$	$U^\dagger U = I$	property $U^\dagger U = I$ and $V V^\dagger = I$ where I is the identity;
1605.05775_F00103	7	■ 1.000	$V V^\dagger = I$	$V V^\dagger = I$	property $U^\dagger U = I$ and $V V^\dagger = I$ where I is the identity;
1605.05775_F00104	7	■ 1.000	I	I	property $U^\dagger U = I$ and $V V^\dagger = I$ where I is the identity;
1605.05775_F00105	7	■ 1.000	m^2	m^2	vector Φ is mapped into a much smaller m^2 dimensional
1605.05775_F00106	7	■ 1.000	$\tilde{\Phi}(\mathbf{x})$	$\tilde{\Phi}(\mathbf{x})$	$\tilde{\Phi}(\mathbf{x})$ with m^2 components as illustrated in Fig. 12(c).
1605.05775_F00107	7	■ 1.000	C^ℓ	C^ℓ	Finally, $f^\ell(\mathbf{x})$ is computed by contracting $\tilde{\Phi}(\mathbf{x})$ with C^ℓ .
1605.05775_F00108	7	■ 1.000	α_i	α_i	V tensors of the MPS Eq. (17) that the indices α_i and
1605.05775_F00109	7	■ 1.000	α_{i+1}	α_{i+1}	α_{i+1} of the core tensor C label an orthonormal basis for
1605.05775_F00110	8	■ 0.896	$U_{\alpha_j s_j}^{\alpha_{j+1}+1}$	$U_{\alpha_j s_j}^{\alpha_{j+1}+1}$	the MPS. For example, the tensor $U_{\alpha_j s_j}^{\alpha_{j+1}+1}$ tensor at site j
1605.05775_F00111	8	■ 1.000	α_{j+1}	α_{j+1}	by α_{j+1} by forming linear combinations of products of
1605.05775_F00112	8	■ 1.000	$ f^\ell(\mathbf{x}) ^2$	$ f^\ell(\mathbf{x}) ^2$	$ f^\ell(\mathbf{x}) ^2$ for each fixed ℓ as a probability distribution func-
1605.05775_F00113	8	■ 1.000	$d\mu(x)$	$d\mu(x)$	require for some integration measure $d\mu(x)$ that the dis-
1605.05775_F00114	8	■ 1.000	$\phi^s(x)$	$\phi^s(x)$	constituent local maps $\phi^s(x)$ have the property
1605.05775_F00115	8	■ 1.000	ϕ^s	ϕ^s	that is, if the components of ϕ^s are orthonormal functions
1605.05775_F00116	8	■ 1.000	$d\mu(x) = 2dx$	$d\mu(x) = 2dx$	(21) if the integration measure chosen to be $d\mu(x) = 2dx$.
1605.05775_F00117	8	■ 1.000	N_s	N_s	W . From this weight tensor we sampled N_s data points
1605.05775_F00118	8	■ 1.000	$\ell = A$	$\ell = A$	1. Sample a label $\ell = A$ or $\ell = B$ according to the
1605.05775_F00119	8	■ 1.000	$\ell = B$	$\ell = B$	1. Sample a label $\ell = A$ or $\ell = B$ according to the
1605.05775_F00120	8	■ 0.984	$P_A = \int_{\mathbf{x}} f^A(\mathbf{x}) ^2 = \sum_{s_1 s_2} W_{s_1 s_2}^A ^2$	$P_A = \int_{\mathbf{x}} f^A(\mathbf{x}) ^2 = \sum_{s_1 s_2} W_{s_1 s_2}^A ^2$	probabilities $P_A = \int_{\mathbf{x}} f^A(\mathbf{x}) ^2 = \sum_{s_1 s_2} W_{s_1 s_2}^A ^2$
1605.05775_F00121	8	■ 1.000	$P_B = 1 - P_A$	$P_B = 1 - P_A$	and $P_B = 1 - P_A$.
1605.05775_F00122	9	■ 1.000	$\sigma / \sqrt{N_s}$	$\sigma / \sqrt{N_s}$	training points N_s . The solid curve is a fit of the form $\sigma / \sqrt{N_s}$.
1605.05775_F00123	9	■ 1.000	$\mathbf{x} = (x_1, x_2)$	$\mathbf{x} = (x_1, x_2)$	2. Sample a data point $\mathbf{x} = (x_1, x_2)$ according to the
1605.05775_F00124	9	■ 1.000	$p(\mathbf{x} \ell) = f^\ell(\mathbf{x}) ^2 / P_\ell$	$p(\mathbf{x} \ell) = f^\ell(\mathbf{x}) ^2 / P_\ell$	distribution $p(\mathbf{x} \ell) = f^\ell(\mathbf{x}) ^2 / P_\ell$ for the chosen ℓ .
1605.05775_F00125	9	■ 0.960	$\tilde{W}$	$\tilde{W}$	toy model with weight tensor $\tilde{W}$ using the log-likelihood
1605.05775_F00126	9	■ 1.000	$p(\ell, \mathbf{x}) = f^\ell(\mathbf{x}) ^2 = W^\ell \cdot \Phi(\mathbf{x}) ^2$	$p(\ell, \mathbf{x}) = f^\ell(\mathbf{x}) ^2 = W^\ell \cdot \Phi(\mathbf{x}) ^2$	where $p(\ell, \mathbf{x}) = f^\ell(\mathbf{x}) ^2 = W^\ell \cdot \Phi(\mathbf{x}) ^2$ and $\tilde{p}(\ell, \mathbf{x})$ has
1605.05775_F00127	9	■ 1.000	$\tilde{p}(\ell, \mathbf{x})$	$\tilde{p}(\ell, \mathbf{x})$	where $p(\ell, \mathbf{x}) = f^\ell(\mathbf{x}) ^2 = W^\ell \cdot \Phi(\mathbf{x}) ^2$ and $\tilde{p}(\ell, \mathbf{x})$ has
1605.05775_F00128	9	■ 0.894	D_{KL}	D_{KL}	age D_{KL} as a function of number of sampled training
1605.05775_F00129	9	■ 1.000	σ	σ	$\sigma / \sqrt{N_s}$ where σ is a fitting parameter. The results indi-
1605.05775_F00130	9	■ 1.000	L_1	L_1	tensor network, such as weight decay or L_1 penalties.

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1605.05775_F00131	10	■ 1.000	$\phi^{s_j}(\mathbf{x}_j) \text{ for } s_j=1,2,\dots,d$	$\phi^{s_j}(x_j), s_j = 1, 2, \dots, d$	component vector $\phi^{s_j}(x_j)$, $s_j = 1, 2, \dots, d$. We always
1605.05775_F00132	10	■ 1.000	$\theta_j = \frac{\pi}{2} x_j$	$\theta_j = \frac{\pi}{2} x_j$	larger d is as follows. Define $\theta_j = \frac{\pi}{2} x_j$. Because
1605.05775_F00133	10	■ 1.000	$\cos^2(\theta_j) + \sin^2(\theta_j) = 1$	$(\cos^2(\theta_j) + \sin^2(\theta_j)) = 1$	$(\cos^2(\theta_j) + \sin^2(\theta_j)) = 1$, then also
1605.05775_F00134	10	■ 0.989	$\binom{n}{k} = n! / (k!(n-k)!)$	$\binom{n}{k} = n! / (k!(n-k)!)$	efficients $\binom{n}{k} = n! / (k!(n-k)!)$.
1605.05775_F00135	10	■ 1.000	$\sum_{s_j} \phi^{s_j}(\mathbf{x}_j) ^2 = 1$	$\sum_{s_j} \phi^{s_j} ^2 = 1$	that $\sum_{s_j} \phi^{s_j} ^2 = 1$ for larger d . (These functions are ac-
1605.05775_F00136	10	■ 1.000	$W^\ell \cdot \Phi(\mathbf{x})$	$W^\ell \cdot \Phi(\mathbf{x})$	product $W^\ell \cdot \Phi(\mathbf{x})$ to realize a richer class of functions.
1605.05775_F00137	10	■ 1.000	$\cos(\frac{\pi}{2} x_j), \cos(\frac{3\pi}{2} x_j), \dots$	$\cos\left(\frac{\pi}{2} x_j\right), \cos\left(\frac{3\pi}{2} x_j\right), \dots$	frequency terms of the form $\cos\left(\frac{\pi}{2} x_j\right), \cos\left(\frac{3\pi}{2} x_j\right), \dots$,
1605.05775_F00138	10	■ 1.000	$\cos(\frac{(d-1)\pi}{2} x_j)$	$\cos\left(\frac{(d-1)\pi}{2} x_j\right)$	$\cos\left(\frac{(d-1)\pi}{2} x_j\right)$ and similar terms replacing cos with sin.